A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 38

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 3 points below. **1 Renewable:** re-forms naturally relatively quickly so use does not necessarily reduce future availability (solar, wind, tidal, geothermal, biofuel) **1 Depletable:** use can reduce future availability - all non-renewables (fossil fuels, uranium) plus renewables exploited unsustainably **1** Wood / biofuels are renewable but depletable if forests are felled faster than they regrow **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	3

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 3 points below. **1** Other factors (technology, location, cost) restrict how much can be harnessed **1 Examples:** coal too deep to mine – winds high above the ground / far offshore where turbines cannot be placed – waves in deep water where equipment cannot be anchored – low-intensity sunlight that cannot produce high temperatures **1** New technologies or higher prices may make these usable in future **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
01.3	**Any 4 of 7** 1 mark each for any 4 of the 7 points below, in any order, maximum 4. **1 Wind:** strong reliable winds - shallow seas, open plains, uplands – low land-use conflict (not near urban areas, ecologically sensitive or scenic areas) – near the grid **1 HEP:** high reliable rainfall, large catchment, narrow valley for a small dam with a large basin, impermeable stable bedrock **1 Solar:** high light intensity, low cloud cover **1 Wave:** coasts with strong reliable prevailing winds and a long fetch **1 Tidal:** large tidal range, coastal features (estuaries, channels) that focus the flow **1 Geothermal:** hot rocks near the surface / recent volcanic activity **1 Biofuels:** nearby farmland / forest / urban waste to avoid transporting bulky low-density fuel **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliable** → repeatable / gives consistent results when repeated • **HEP** → electricity generated from falling water • **impermeable** → does not let water through **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-177; AQA spec 3.3.2.	
02.1	**1 mark each** 1 mark for each of the 3 points below. **1** Clay particles are very small and carry electrostatic charges on their surfaces so they repel each other and stay suspended **1** Flocculants (aluminium sulfate / alum or polyelectrolytes) are added and mixed to neutralise the charges **1** The particles clump together (flocs) and settle in the clarifier tank **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** Increases dissolved oxygen and removes hydrogen sulfide (bad-egg smell) from anaerobic sources **1** Oxidises some dissolved metals (eg iron) so they become insoluble and can be removed – improves taste **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** Peaty water contains organic matter which reacts with chlorine to produce toxic substances (eg trihalomethanes) **1** Ozone / UV kill pathogens without forming these by-products (though chlorine / chloramine keeps water sterile during distribution) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.4	**1 mark each** 1 mark for each of the 3 points below. **1 Ion exchange:** toxic ions adsorb onto polymer resin beads or zeolite (also used to soften water by removing calcium / magnesium) **1** Reverse osmosis / distillation (desalination) - expensive, energy-intensive **1 Activated carbon:** organic molecules causing odour / taste and pesticides adsorb onto the high-surface-area carbon particles **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **adsorb** → nutrient ions sticking to the surfaces of clay and humus **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 126-128; AQA spec 3.2.2.4.	

Do not write outside the box

Question	Marking guidance	Marks
03.1	**1 mark each** 1 mark for each of the 5 points below. **1** Use a calibrated sound meter (dB(A)) at fixed distances (eg 10, 50, 100, 250, 500 m) from the track behind each barrier and at a control site with no barrier **1 Standardise the noise source:** same cars / same race or a standard test noise – same time of day, wind speed / direction, weather – same height of meter above ground **1** Take repeated readings at each distance and calculate means (and standard deviation) to assess variability **1** Use a measure such as LA10 or Leq over a set period rather than a single peak reading **1** Compare reduction in dB relative to the control at each distance – plot noise level against distance for each barrier – statistical test (t-test) to assess significance **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **standard deviation** → a measure of the spread of the data about the mean • **t-test** → a test for a difference between the means of two normally distributed sets **Do not credit** • 'the result was significant' without comparing the calculated value with the critical value at p = 0.05	5
03.2	1 mark for the point below. **1** Wind carries sound / affects how far it travels so different wind directions would change readings independently of the barrier **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** LA10 is the noise level exceeded for 10% of the period - a measure of the noisiest periods **1** LA90 is the level exceeded for 90% of the period - a measure of the quietest / background level **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2

Do not write outside the box

Question	Marking guidance	Marks
03.4	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Double / triple glazing **1** Acoustic insulation in walls / roof **1** Bedrooms on the side facing away from the track **1** Dense hedges / trees and earth mounds in gardens **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 293-295, 418; AQA spec 3.4.3.2.13.	
04.1	**1 mark each** 1 mark for each of the 3 points below. **1** Net held with its straight edge on the river bed facing the current (open to the flow) **1** The bed in front is disturbed by kicking / scraping a boot for a standard number of kicks and time (eg 30 s) **1** Same substrate type, water depth and flow rate at both sites – same net mesh – repeat samples – invertebrates identified with a key and counted **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **kick sampling** → disturbing the stream bed and catching the organisms carried downstream	3
04.2	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Difficult to standardise the vigour / area kicked **1** Buried organisms or those fixed to stones are not dislodged **1** Some organisms swim and avoid the net **1** Flow rate varies and affects capture **1** Fragile organisms destroyed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **kick sampling** → disturbing the stream bed and catching the organisms carried downstream **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3

Do not write outside the box

Question	Marking guidance	Marks
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** A fixed area of bed is enclosed by a frame so the sampled area is standardised – side nets reduce escapes **1** But only works in fast enough flow and is hard to set on rough substrates / deep water **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **kick sampling** → disturbing the stream bed and catching the organisms carried downstream	2
04.4	**1 mark each** 1 mark for each of the 2 points below. **1** Pollutant / oxygen levels fluctuate (eg with flow, time of day) so one reading may miss a pollution event **1** Sensitive taxa (stonefly, mayfly larvae) disappear and tolerant ones (midge larvae, hoglouse) remain, integrating conditions over weeks / months (biotic index) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biotic** → living factors / effects of other organisms	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 395-396, 415-416; AQA spec 3.7.2.1.	
05.1	**1 mark each** 1 mark for each of the 2 points below. **1** Predicting the route / movement a pollutant will take through the environment (based on its properties and features of the environment) **1** To assess where it will become concentrated / which locations and organisms are most at risk, so monitoring can focus on those sites **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
05.2	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Sea current velocity and direction **1 Properties of the radionuclides:** half-life, solubility, tendency to adsorb onto sediments / bioaccumulate **1** Food chain pathways, eg uptake by seaweed, shellfish, fish eaten by people **1** Geology / hydrology of the coast – sediment deposition areas (mud flats) **1 Human activities in the area:** fishing, seaweed collection, beach use **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **half-life** → the time for the activity, or the amount, to fall by half • **adsorb** → nutrient ions sticking to the surfaces of clay and humus **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'after two half-lives nothing is left' - each half-life halves what remains, so some activity always continues • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	3
05.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Identify the group of people whose lifestyle gives them the highest exposure (eg people who eat a lot of locally caught shellfish / fish, live near the mud flats, collect seaweed) **1** Monitor their exposure / the concentration of radioactive material in the foods they eat or their body **1** If their exposure / risk is acceptably low, everyone else is assumed to be at even lower risk – if not, emissions are controlled **1** CGM assesses exposure before health effects occur (not used to detect damage) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Critical Group** → the people most at risk / most heavily exposed	3

Do not write outside the box

Question	Marking guidance	Marks
05.4	**Any 2 of 7** 1 mark each for any 2 of the 7 points below, in any order, maximum 2. **1** Seawater **1** Sediment / mud **1** Seaweed **1** Shellfish / molluscs / fish **1** Milk, vegetables or meat from nearby farms **1** Atmospheric dust **1** Soil **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243, 303-304; AQA spec 3.4.3.1.	
06.1	**1 mark each** 1 mark for each of the 4 points below. **1** CO_2 rises from about 412 ppm in January to a peak of 416 ppm in May, falls to a minimum of 407 ppm in September, then rises again **1** During spring / summer, photosynthesis by plants (growing season) removes CO_2 faster than respiration releases it, so the concentration falls **1** In autumn / winter photosynthesis is reduced (leaf fall) but respiration and decomposition continue, so CO_2 rises **1** The atmosphere is in dynamic equilibrium: the concentration fluctuates around a mean – more land / vegetation in the northern hemisphere makes the cycle large **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Natural processes that add gases (respiration, combustion, volcanoes, denitrification) are balanced by processes that remove them (photosynthesis, dissolving, nitrogen fixation) **1** So the average composition stays the same over long timescales although the rates vary and concentrations fluctuate around a mean **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **denitrification** → nitrate converted back to nitrogen gas • **nitrogen fixation** → nitrogen gas converted into ammonium or other nitrogen compounds	2

Do not write outside the box

Question	Marking guidance	Marks
06.3	**Any 2 of 5** 1 mark each for any 2 of the 5 points below, in any order, maximum 2. **1** Absorbs harmful UV / solar wind radiation in the stratosphere (ozone layer) **1** Greenhouse gases delay the escape of infrared, keeping the surface warm (33 °C warmer than without) **1** Atmospheric pressure allows liquid water to exist **1** Distributes heat from the tropics to higher latitudes by winds **1** Transports water vapour to areas that would otherwise be dry **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ozone layer** → the ozone in the stratosphere that absorbs UV • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	2
06.4	**1 mark each** 1 mark for each of the 2 points below. **1** Troposphere **1** Stratosphere **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-89; AQA spec 3.2.1.1.	
07.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (2900 - 1400) / 1400 x 100 **1** = 107% (accept 107.1) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** Large decrease from 24 per year (1975-79) to 2 per year (2010-14) **1** Steepest fall between the late 1970s and 1980s – slower decline / levelling off after 1995 **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.3	**Any 3 of 6** 1 mark each for any 3 of the 6 points below, in any order, maximum 3. **1** Double-hulled tankers required for all large tankers **1** Improved navigation (GPS, AIS) and offshore shipping routes **1** Inert gas systems reducing explosions **1** Twin engines / rudders reducing groundings after mechanical failure **1** Stricter regulation, inspection and polluter-pays liability (eg after Exxon Valdez) **1** Separate ballast tanks / crude oil washing **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
07.4	**1 mark each** 1 mark for each of the 3 points below. **1** Most oil pollution comes from dispersed / diffuse sources not tanker accidents: waste lubricating oil, industrial oil in drains, road runoff **1 Routine operations:** small spills, pipeline leaks, storage tank leakage, oil rig accidents (eg Deepwater Horizon) **1** Growth in oil use and transport means more small releases even if large accidents are rarer **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **diffuse** → pollution from a wide area with no single outlet	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 258-262; AQA spec 3.4.3.2.7.	
08.1	1 mark for the point below. **1** The length of time a pollutant remains in the environment before it is broken down / degraded (chemically or biologically) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistence** → does not break down / stays in the environment for a long time	1

Do not write outside the box

Question	Marking guidance	Marks
08.2	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1 Light:** UV light degrades many pesticides (eg neonicotinoids break down fastest in sunlight) so persistence is shorter in bright conditions **1 Temperature:** higher temperatures increase the rate of chemical breakdown / microbial decomposition **1 Soil microbes / oxygen:** aerobic microbial activity breaks pesticides down so persistence is shorter in well-aerated, biologically active soils **1** pH / presence of other chemicals may increase or decrease the rate of degradation **1** Adsorption onto clay / organic matter may protect from degradation and increase persistence **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistence** → does not break down / stays in the environment for a long time • **Adsorption** → nutrient ions sticking to the surfaces of clay and humus **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-266; AQA spec 3.4.3.2.8.	
09.1	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Public opposition following accidents (Three Mile Island 1979, Chernobyl 1986, Fukushima 2011) **1** Very complex, expensive technology needing an advanced industrial infrastructure **1** Concerns about links between civil nuclear materials and weapons / terrorism **1** Uncertainty over permanent disposal of radioactive waste **1** Uncertain decommissioning costs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **decommissioning** → safely dismantling a power station at the end of its life **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	3

Do not write outside the box

Question	Marking guidance	Marks
09.2	**1 mark each** 1 mark for each of the 3 points below. **1** Only 0.7% of natural uranium is fissile U-235 – the remaining 99.3% is non-fissile U-238 (a 'fertile' fuel) **1** Neutron bombardment in the reactor converts U-238 to fissile plutonium-239 (via beta decays) **1** So much more energy is harnessed from the mined uranium – fast reactors need no moderator but are complex and costly **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **moderator** → slows the neutrons so that fission continues	3
09.3	**2 marks per point** 2 marks for each of the 2 points below; award 1 mark where a point is only partly made. **2 Advantages:** thorium is three times more abundant – much less radioactive waste with shorter half-lives – harder to make weapons material – no fuel enrichment needed **2 Disadvantages:** thorium-232 is not fissile so must be bred into uranium-233 (slow breeding rate) – U-233 is an alpha emitter so very hazardous to handle – less developed technology with high development costs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 192-194; AQA spec 3.3.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Control of greenhouse gases:** energy conservation (transport, buildings, industry), renewable and nuclear energy replacing fossil fuels, CCS, reduced deforestation and afforestation (carbon sequestration), methane capture from landfill / digesters, HFC controls, changes in agriculture • **Geoengineering:** white roofs, ocean fertilisation, solar shades - largely untried, possible unpredictable consequences • **International agreements:** Kyoto Protocol, Paris Agreement - targets, but non-binding / uneven participation – carbon trading and taxes – comparison of per capita carbon footprints between countries • **Evidence of limited success:** global CO2 concentrations still rising (about 280 ppm pre-industrial to over 410 ppm) – temperatures up about 1 °C – emissions in some developed countries falling but rising in developing economies • **Reasons for difficulty:** fossil fuels underpin economies, need for major lifestyle change, cost, time delays and natural fluctuations masking effects, political disagreement, developing countries' right to develop • Adaptation strategies (flood defences, drainage, managed retreat, infrastructure changes) reduce impacts but do not address the cause • Comparison with the Montreal Protocol's success for ozone • **Judgement:** partial success - technology exists and some regional reductions achieved, but global strategies have not yet stopped the rise – tipping points make urgency greater **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	9
	• Any accurate, relevant material that is not in the indicative content. • **energy conservation** → using less energy, eg by insulation or behaviour change • **CCS** → capturing carbon dioxide from flue gases and storing it underground **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	

Do not write outside the box

Question	Marking guidance	Marks
	Model answer - Level 3 (8 marks)	
	Strategies fall into control of greenhouse gases, geoengineering, international agreements and adaptation. Emissions are cut by energy conservation in transport, buildings and industry, by replacing fossil fuels with renewables and nuclear, by carbon capture and storage, by reducing deforestation and planting forests, and by capturing methane from landfill. Geoengineering ideas - white roofs, ocean fertilisation, solar shades - remain largely untried with unpredictable side effects. Adaptation, such as flood defences and managed retreat, reduces impacts but not the cause.	
	The Kyoto Protocol and Paris Agreement set targets, but they are weak or non-binding and participation is uneven; carbon trading and taxes exist but per capita footprints still differ enormously between countries. The evidence of success is limited: atmospheric CO_2 has risen from about 280 ppm before industrialisation to over 410 ppm and temperature by about 1 °C, and although emissions are falling in some developed countries they are rising in developing economies.	
	The difficulty is that fossil fuels underpin economies, so control demands lifestyle change, cost, and political agreement between nations with different priorities, while time delays and natural fluctuations mask the effect of any action. By contrast the Montreal Protocol succeeded because ozone was a substitution problem. The judgement must be partial success: the technology exists and regional reductions prove it works, but global strategies have not yet stopped the rise, and tipping points make the delay dangerous.	
	Weaker answer - Level 2 (5 marks)	
	Methods to reduce climate change include using renewable energy such as wind and solar instead of fossil fuels, saving energy, and planting trees to absorb carbon dioxide. International agreements like the Kyoto Protocol and the Paris Agreement have set targets for countries to reduce emissions. However carbon dioxide levels are still rising, so the methods have not been very successful so far. This is because countries rely on fossil fuels and do not always keep to the agreements. More needs to be done.	
	What makes the difference	
	• Level 2 covers the main strategies and reaches a conclusion, but with no quantitative evidence (280 to 410 ppm, 1 °C) and no mention of geoengineering or adaptation.	
	• Level 3 explains *why* success is limited (economic dependence, time lags, natural variability, political disagreement) rather than just stating that it is.	
	• The Montreal comparison and the tipping-point point give Level 3 the breadth and judgement the top band requires.	
10.2	1 mark for the point below. **1** The total greenhouse gas / CO2 emissions caused by an individual, organisation, product or country (often expressed per capita per year) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 89-107; AQA spec 3.2.1.2.	
11.1	**Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Composition and structure:** N_2, O_2, Ar, CO_2, ozone – troposphere and stratosphere – pressure and liquid water – gases for biological molecules and industry • **Energy processes:** UV absorbed by O / O_2 / O_3 dynamic equilibrium – visible light absorbed at the surface – infrared absorbed by greenhouse gases (33 °C warming) – heat distribution by winds and ocean currents – transport of water vapour – energy budget and albedo • **Global climate change:** anthropogenic greenhouse gases, sources, feedback and tipping points, consequences for climate, sea level, currents, species – difficulties of monitoring and prediction – control strategies and agreements • **Ozone depletion:** CFCs and halons, catalytic destruction, effects of UV, Montreal Protocol and replacements, recovery time • **Other threats:** aerosols, smoke, land-use change – interconnection of processes	25
	• **Evaluation:** which functions are most at risk – success of ozone controls vs slow progress on greenhouse gases – justified conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-113; AQA spec 3.2.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss how the design of a sampling programme and the choice of statistical analysis affect the validity of conclusions drawn from environmental investigations.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Attributes:** accuracy, precision, representative samples, reliability, anomalies, validity – hypothesis and null hypothesis – preliminary studies • **Sampling design:** random vs systematic location and avoiding bias – timing (diurnal, seasonal, long-term, weather) – sample size and number and the running mean – standardisation of technique and control of other variables • Techniques for abiotic and biotic factors and their limitations (quadrats, transects, traps, mark-release-recapture assumptions, keys, meters and calibration) • **Statistics:** variability, standard deviation and error bars, significance and p-values, correlation vs causation – choosing Spearman's rank, chi-squared, t-test, Mann-Whitney U • **Application to Paper 1 topics:** monitoring pollution (noise, air quality, water), climate proxies, energy surveys, soil analysis • Evaluation of trade-offs (cost, time, disturbance) and justified conclusion	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393, 420-423; AQA spec 3.7.	

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